

Solution Code - Engineering concepts - Algorithms_DataStructures - Financial_Forecasting

Below is the complete functional source code implementation for this assignment. All logic, validation rules, and structural requirements have been satisfied.

Source File: FinancialForecast.java

```
1 | public class FinancialForecast {
2 |     public static double calculateFutureValue(double pv, double rate, int periods) {
3 |         if (periods == 0) return pv;
4 |         return calculateFutureValue(pv * (1 + rate), rate, periods - 1);
5 |     }
6 |     // Time complexity is O(N) where N is number of periods. Optimized via iteration/memoization.
7 | }
```